Report

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1 1. Introduction

The US cell phone service market is highly saturated, with most adults already subscribed to a provider. As a result, telecom companies can't rely on steady growth from new customers. Instead, these companies depend mostly on retaining their existing user/customer base. Losing a customer means losing future monthly revenue and often incurring additional costs to acquire replacements. This makes the ability to identify customers who are likely to leave strategically important for profitability, scale, and efficiency.

This project analyzes the IBM Telco Customer Churn dataset, which simulates a telecom company's customer base. Each row represents a single customer and includes demographic information, subscribed services, billing details, and whether the customer left in the past month. The goal of this analysis is to identify patterns associated with customer churn and to evaluate two predictive models (logistic regression and a pruned decision tree) to assess how well they identify high-risk customers. The analysis is exploratory and focuses on extracting insights that could inform realistic retention strategies.

This analysis uses the IBM Telco Customer Churn dataset \cite{blastchar2018telco}.

2 2. Background and Motivation

Telecom companies operate in a highly competitive environment with limited opportunities for growth through new customers. This means retaining existing customers is actually usually more cost effective than continually attracting new ones. Predictive churn modeling offers a way to identify customers who are at higher risk of leaving, allowing companies to intervene early through targeted support, pricing adjustments, or retention incentives.

This project has 3 main questions:

Are newer customers with shorter tenure more likely to churn?

How are pricing and contract structure related to churn risk?

How well can two simple models (logistic regression and a pruned decision tree) distinguish churners from non-churners?

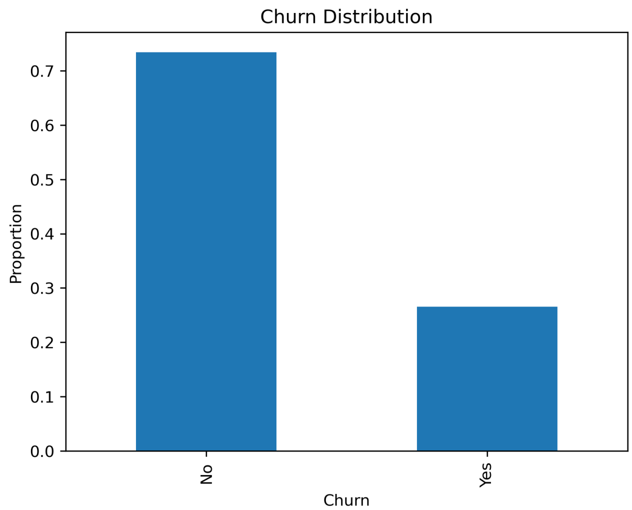
The IBM Telco Customer Churn dataset includes demographic attributes, subscribed services, contract type, billing information, and an indicator of whether the customer churned in the past month. Churn is recorded as a binary outcome, where Yes indicates a customer left and No indicates they stayed. Around 3/4th of customers in the dataset do not churn, creating a class imbalance that complicates model evaluation. Because of this imbalance, accuracy alone can be misleading, so we use AUC (Area Under the ROC Curve) as a complementary metric to assess model performance.

The dataset contains several categories of variables, including demographic characteristics such as gender and senior citizen status, service related features such as phone plans, internet service, and streaming options, and account information such as tenure, contract type, monthly charges, payment method, and total charges. Together, these features allow us to explore whether newer customers are more likely to leave, whether higher monthly bills are associated with churn, and whether longer contract commitments are linked to greater customer retention. Throughout the analysis, we assume that churn labels are accurate, services are correctly recorded, and that there is no systematic measurement bias in the billing or tenure variables.

3 3. Exploratory Data Analysis and Key Observations

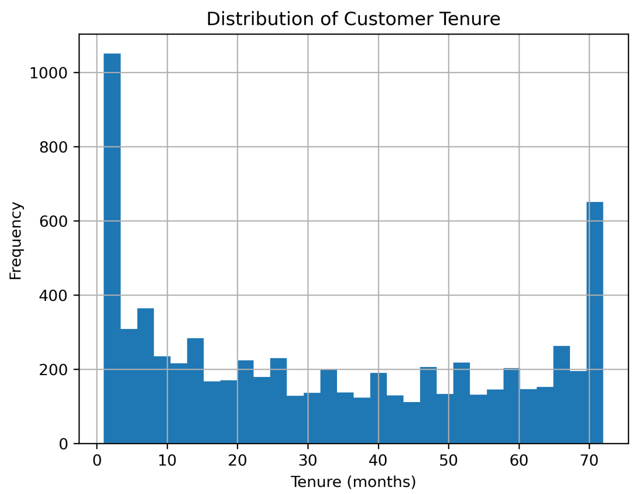
Before modeling, we examined the distributions and relationships among the variables. Below are the most important visualizations used for our project's goals.

3.1 Figure 1. Churn Distribution



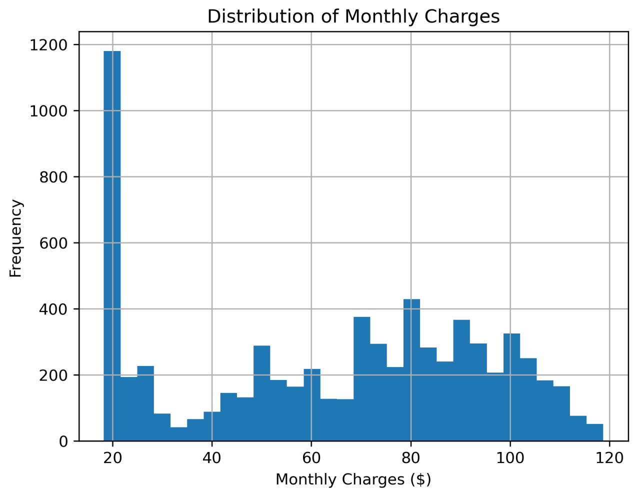
Caption:
Figure 1 shows that around ~75% of customers do not churn. This imbalance makes accuracy alone an inadequate performance metric, motivating the use of AUC.

3.2 Figure 2. Distribution of Customer Tenure



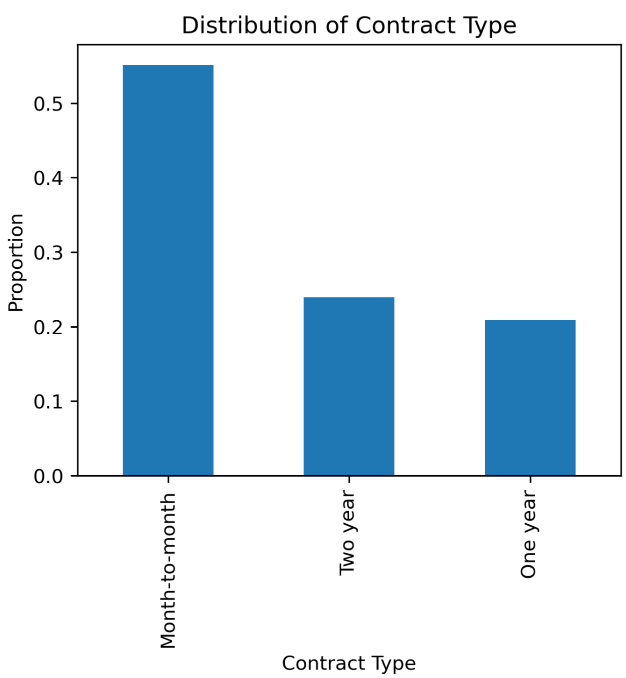
Caption:
Figure 2 shows the distribution of tenure. Many customers are either very new or near the maximum tenure of 72 months, suggesting two distinct groups in the customer base (New and Long Standing Customer).

3.3 Figure 3. Distribution of Monthly Charges



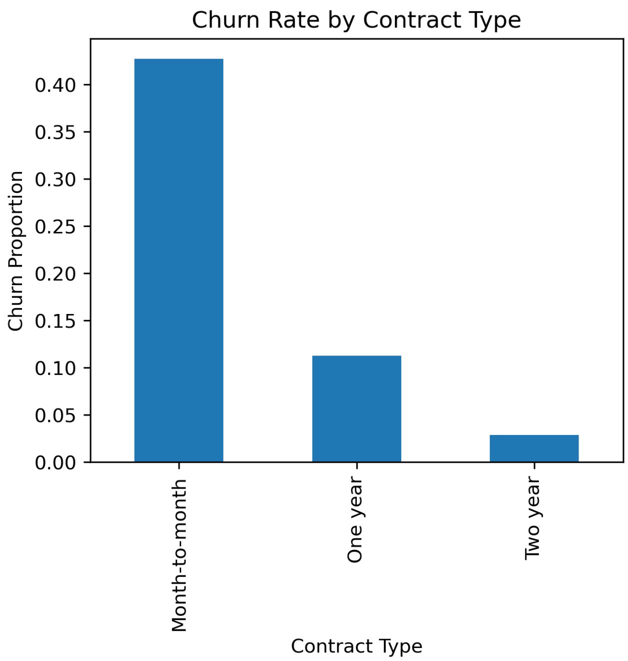
Caption:
Figure 3 shows large variation in monthly charges, showing differences in service bundles and pricing plans across customers.

3.4 Figure 4. Contract Type Distribution



Caption:
Figure 4 shows that most customers are on month-to-month contracts which often provide less commitment and greater likelihood of churn.

3.5 Figure 5. Churn Rate by Contract Type



Caption:
Figure 5 shows that churn rates are much higher among month-to-month customers (matching reasoning), while one-year and two-year contract customers churn far less frequently. Contract type appears to be one of the strongest predictors of churn.

4 4. Modeling Approach

Two predictive models were used in this analysis: logistic regression and a decision tree classifier. These models are transparent, easy to interpret, and consistent with the exploratory goals of the project.

4.0.1 Logistic Regression

We trained a logistic regression model using all features except customerID and the churn label. Logistic regression provides a simple and commonly used baseline for churn prediction and outputs estimated probabilities of churn. All categorical variables were converted into indicator variables prior to modeling.

The data were split into a training set and a 20 percent held out test set using stratified sampling to preserve the churn proportion. Model performance was evaluated using both test accuracy and the area under the ROC curve (AUC), since accuracy alone can be misleading in the presence of class imbalance.

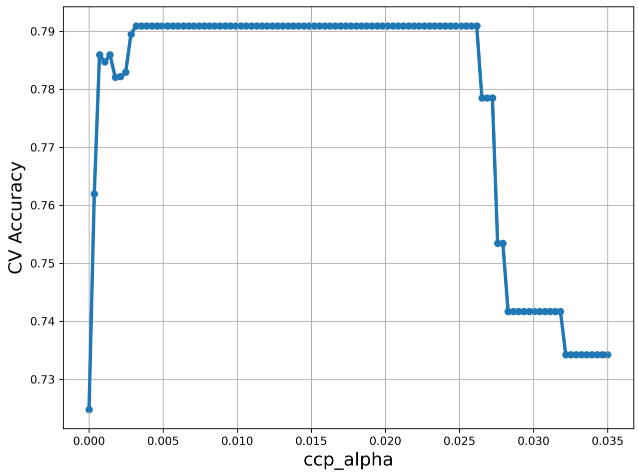
4.0.2 Decision Tree

We also trained a classification tree using the same feature set, again excluding customerID and the churn label. To reduce overfitting, we applied cost-complexity pruning and focused on tuning the pruning parameter ccp_alpha. All other hyperparameters were left at their default values. We performed 5-fold cross-validation over a grid of 100 candidate ccp_alpha values ranging from 0 to 0.035, using accuracy as the scoring metric. The cross-validation results had accuracy increase from very small values of ccp_alpha, remained relatively stable across a broad middle range, and then declined once the tree became over-pruned. Based on that, we chose an optimal value of ccp_alpha = 0.0032. The resulting pruned tree achieved a test accuracy of 0.7818. Decision trees offer greater flexibility than logistic regression by capturing nonlinear interactions, but this comes at the cost of higher variance, making pruning an important step.

4.0.3 Model Comparison

To compare the two models, we computed both accuracy and AUC on the test set. The logistic regression model outperformed the decision tree on both metrics. While accuracy measures the overall fraction of correctly classified customers, AUC captures how well a model ranks churners above non-churners across all probability thresholds, which we already mentioned but is important for imbalanced outcomes like churn.

4.1 Figure 6. Cross-Validation Accuracy vs ccp_alpha



Caption:
Figure 6 shows cross validated accuracy for different pruning strengths. Accuracy increases at low levels of pruning, stabilizes across a broad plateau, and declines once the tree is over-pruned. The selected value is ccp_alpha = 0.0032.

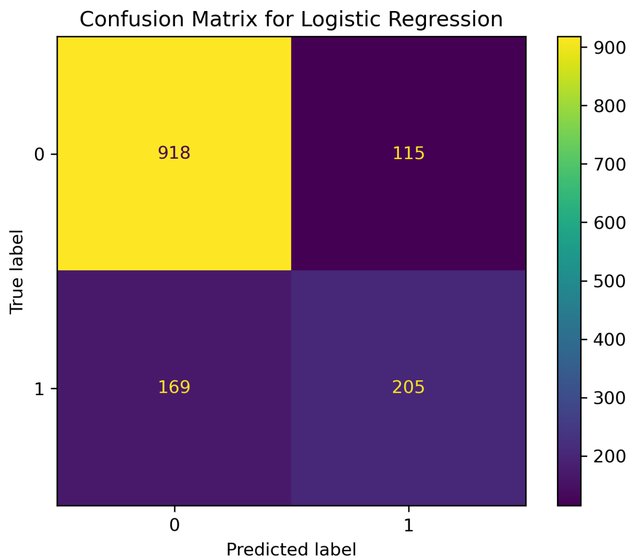
5 5. Results

Logistic regression achieved a test accuracy of about 0.7967 and an AUC of about 0.8305. The confusion matrix shows that logistic regression correctly identifies many non-churners while also catching a meaningful portion of churners (see Figure 3), although some churners are still missed.

The decision tree achieved a test accuracy of about 0.7818 and an AUC of about 0.7937. It correctly classifies many non-churners and produces fewer false positives than logistic regression, but it misses more true churners. This makes the tree more conservative. While its accuracy is only slightly lower than that of logistic regression, its lower AUC and higher number of missed churners reduce its usefulness for retention strategies.

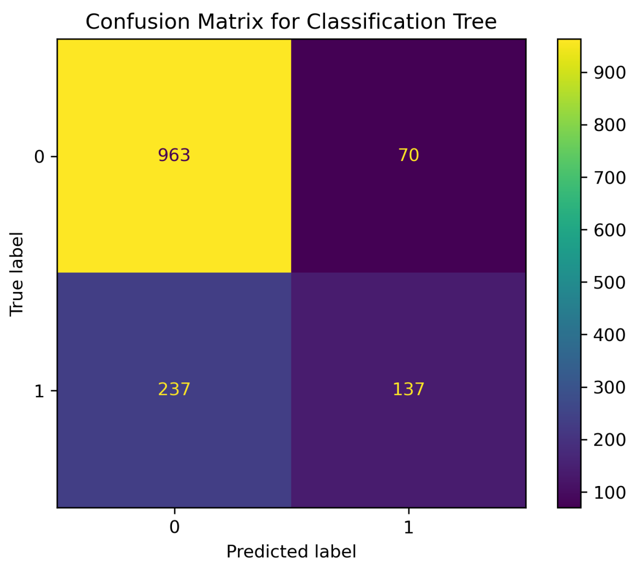
Overall, the logistic regression model performs better on this dataset. It is more balanced across the two classes and identifies more churners, which is often the most important outcome for a business seeking to prevent customer loss.

5.1 Figure 7. Confusion Matrix for Logistic Regression



Caption:
Figure 7 shows that logistic regression correctly classifies most non-churners and captures a meaningful proportion of churners, though some are missed.

5.2 Figure 8. Confusion Matrix for Decision Tree



Caption:
Figure 8 shows that the pruned classification tree produces fewer false positives

but misses more true churners, making it less useful for retention targeting despite similar accuracy.

6 6. Interpretation and Implications

Insights from EDA/predictive modeling:

Customers with short tenure are way more likely to leave, which has an implication that the early months of the relationship are especially fragile/the most important to foster. Therefore, these early months may benefit from focused onboarding and support. This is the time when customers don't have brand loyalty and need good impressions to decrease churmn. Customers with higher monthly charges also churn at higher rates, pointing toward the importance of perceived value. High bills without matching perceived benefits can push people away. Contract type is one of the strongest predictors; month-to-month customers churn far more often than those on one-year or two-year contracts, which lines up with the idea that commitment mechanisms and incentives to stay reduce the likelihood of leaving (locked into a contract with only a few, sparse oppurtunities to leave versus having the choice every single month). Finally, service bundles that add streaming, security, or technical support appear to be associated with lower churn in several of our summaries, which is consistent with the notion that deeper integration into the company's ecosystem or higher percieved value of services makes it harder or less attractive to switch providers.

If a real telecom company observed patterns like these in its own data:

Extra attention could be directed toward customers in their first year through stronger onboarding and early support. This could include things such as clearer explanations of billing and plan features at signup, proactive follow-up emails or messages in the first few months, dedicated customer support queues/priorities for new customers, or small welcome incentives such as temporary discounts, free add-ons, or data boosts. Usage monitoring during the first year could help identify early warning signs such as repeated service issues, unusually low engagement, or frequent customer service contacts, triggering outreach before dissatisfaction escalates. Training customer service representatives to prioritize retention during early interactions, rather than purely issue resolution, may also help stabilize customers before loyalty is established.

Customers with unusually high monthly charges could be flagged for closer review, companies might use internal thresholds or percentile cutoffs to identify customers (through further modeling) with above average charges and then check whether their plans align with actual usage. Interventions could include plan optimization recommendations, clearer communication about what customers are paying for, loyalty discounts, or targeted promotions that reduce bill shock without broadly lowering prices. High charge customers could also be routed to more experienced support staff or offered service guarantees, faster issue resolution, or priority customer service to reinforce value perception. Internally, sales and support teams could be trained to avoid overselling plans that customers do not fully use, which may reduce downstream churn driven by dissatisfaction (less money initally but more money in the long run).

Since month-to-month customers churn far more often, companies could encourage longer commitments by offering incentives such as reduced monthly rates, waived fees, device upgrades, or bundled add-ons for customers who voluntarily move to 1 year or 2 year contracts. Sales representatives could be trained to emphasize the long term cost savings and stability of longer contracts. Companies might also time contract offers strategically, such as after successful service interactions or once customers pass the high risk early tenure period, when customers may be more open to committing. Firms could design bundles that are easy to understand, transparently priced, and clearly valuable rather than having a sneaky business practice of hidden feeds (which may increase revenue but decrease brand perception). Bundles could be customized based on customer profiles, usage patterns, or household characteristics, making them feel more relevant and less like generic add-ons. None of these proposals guarantee any reduced churn, and none should be interpreted as causal prescriptions, but all seem like reasonable suggestions given the patterns found in this dataset.

7 7. Assumptions and Limitations

This analysis relies on several assumptions about the dataset. We assume that each row corresponds to a unique customer account, that churn labels correctly reflect whether the customer left during the period of interest, and that key variables such as tenure, charges, contract type, and service indicators are measured without systematic error. We assume that customers' observations are independent and that no predictor variables directly encode the churn outcome in a way that would trivially reveal the label. We also assume that dropping rows with missing TotalCharges after converting that column to numeric does not introduce bias that fundamentally alters the observed relationships. Finally, we treat the dataset as a reasonable snapshot of customer behavior during a typical period at the fictional company, assuming that the patterns observed are not driven by a short term shock or an unusual event/time.

As for limitations, the data is observational, so we cannot just interpret any of the associations as causal. An example of this is, while customers on month-to-month contracts churn more often, we can't conclude that simply switching a given customer to a longer contract will cause them to stay. Another limitation is that the outcome is imbalanced, with most customers not churning, which makes accuracy alone a weak measure of performance and motivates the use of AUC and confusion matrices to understand how errors are distributed. The dataset is static and does not capture how real telecom markets evolve over time, so models trained here would need to be retrained regularly in any real deployment. One last final limitation is that we only considered two simple models (logistic regression and a pruned decision tree). More complex models might reach higher predictive performance but at the cost of interpretability, computational simplicity, and ease of reproduction. Given the goals of this project and the constraints of the assignment, we chose to prioritize transparency/reproducibility over marginaly increases in accuracy.

8 8. Reproducibility and Workflow

To support reproducibility, utility functions such as compute_accuracy and compute_auc were placed in a separate module (utils.py) with accompanying docstrings and tests. These functions were imported into the analysis notebook to enforce modularity.

The repository also includes:
-An environment.yml file specifying the full software environment
-A Makefile that automates environment setup and notebook execution
-Saved figures written to disk and reused in the main narrative notebook
-PDF builds of notebooks generated using MyST
-A BibTeX file containing all references cited in the report

9 9. Conclusions

This project shows that simple and interpretable models can still uncover meaningful patterns in customer churn data. Both logistic regression and a pruned decision tree identify clear relationships between churn and features such as tenure contract type and monthly charges with logistic regression performing slightly better overall. These results align with common expectations in the telecom industry and suggest that even basic models can provide useful signal when the goal is understanding broad risk patterns rather than maximizing predictive accuracy. However, the findings should be interpreted with caution. This analysis is only exploratory and based on observational simulated data so the results describe associations rather than causal effects. Many real world factors that influence churn are not captured in the dataset and the models are not intended for direct deployment. Still we're able to show how reproducible analysis and straightforward modeling can turn raw data into actionable insight while also making clear where additional data and more advanced methods would be needed for real industry use.

10 Author Contributions

- Marcel Gunadi: Conducted all the exploratory data analysis and built and assessed both models, utils.py
- Benson Chang: utils functions docstring, Main Writeup, README, references file and output figures
- Ethan Chant: Created inital files (repo, website, notebooks), README, Main Writeup, repo orginization, License

- Sophie Hanson: MyST website, PDF exports, Binder badge, environment.yml, Makefile